Air Quality Prediction System

Project Report

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Acknowledgement

The project **"Air Quality Prediction System"** is the Project work carried out by

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Abstract

The Air Quality Prediction System is designed to analyze and forecast air pollution levels using machine learning techniques. With the increasing levels of air pollution in urban areas, it has become essential to monitor and predict air quality in advance to take preventive measures.

This system collects real-time air quality data and historical pollution data using external APIs. The collected data includes parameters such as PM2.5, PM10, nitrogen dioxide (NO_2), sulfur dioxide (SO_2), carbon monoxide (CO), temperature, and humidity. The system processes this data and uses machine learning models such as Random Forest and Long Short-Term Memory (LSTM) to predict the Air Quality Index (AQI) for the upcoming day.

The application also stores the collected data and prediction results in a MySQL database for efficient data management and future analysis. Additionally, a user-friendly dashboard is developed using Streamlit to display current air quality, historical trends, and predicted AQI values in an interactive format.

The proposed system helps users and authorities make informed decisions by providing accurate and timely air quality predictions, thereby contributing to better environmental awareness and public health management.

Introduction

Air pollution has emerged as one of the most critical environmental and public health challenges in recent years. Rapid urbanization, industrial growth, increasing vehicular emissions, and population expansion have significantly contributed to the degradation of air quality, especially in metropolitan cities such as Mumbai and Delhi. Poor air quality not only affects the environment but also has severe impacts on human health, leading to respiratory diseases, cardiovascular problems, and reduced life expectancy.

The Air Quality Index (AQI) is a standardized measure used to evaluate and communicate the level of air pollution in a particular area. It is calculated based on various pollutants such as particulate matter (PM2.5 and PM10), nitrogen dioxide (NO_2), sulfur dioxide (SO_2), carbon monoxide (CO), and ozone (O_3). Monitoring AQI helps individuals understand the quality of air they are exposed to; however, traditional monitoring systems mainly provide real-time data and lack the capability to predict future air quality conditions.

In order to overcome this limitation, there is a growing need for intelligent systems that can analyze historical data and forecast AQI values in advance. Predicting air quality can help individuals, healthcare organizations, and government authorities take timely preventive actions, such as issuing alerts, reducing outdoor activities, or implementing pollution control measures.

The proposed Air Quality Prediction System addresses this need by combining data analysis, machine learning, and real-time data integration. The system collects air pollution data from external APIs, including both current and historical information. This data is then preprocessed to handle missing values, normalize features, and prepare it for model training and prediction.

Machine learning models such as Random Forest and Long Short-Term Memory (LSTM) are used to analyze patterns in the data and predict future AQI values. Random Forest is effective for handling structured data and capturing complex relationships, while LSTM is well-suited for time-series forecasting, making it ideal for predicting AQI based on previous trends.

In addition to prediction, the system includes a user-friendly dashboard developed using Streamlit. This dashboard provides real-time AQI values, historical data trends, and predicted AQI results in a visually interactive format. The integration of a MySQL database ensures that all data, including pollution parameters and prediction results, are stored in a structured and efficient manner for future analysis.

Overall, the Air Quality Prediction System aims to provide an intelligent, scalable, and user-centric solution for monitoring and forecasting air quality. By enabling early detection of pollution levels, the system helps in raising awareness and supports informed decision-making for better environmental management and public health protection.

System Analysis

System analysis is a crucial phase in the development of the Air Quality Prediction System. It involves understanding the problem of increasing air pollution and the need for predicting air quality in advance. This phase focuses on analyzing user requirements, studying existing air quality monitoring systems, and identifying their limitations. It also defines the functional and non-functional requirements of the proposed system, which aims to provide accurate AQI predictions using machine learning techniques along with real-time data visualization.

1.Problem Definition

Air pollution in urban areas has increased significantly due to industrial activities, vehicle emissions, and population growth. Existing air quality monitoring systems mainly provide real-time data but lack the ability to predict future air quality conditions.

Users, including the general public and authorities, require a system that not only displays current air quality but also provides future predictions so that necessary preventive actions can be taken. Therefore, there is a need for an intelligent system that can analyze historical data and forecast AQI accurately.

2. Objectives of the System

The main objective of the Air Quality Prediction System is to develop an intelligent and efficient solution for monitoring and predicting air pollution levels. The system aims to analyze historical and real-time data to provide accurate Air Quality Index (AQI) predictions, helping users take preventive actions.

- To collect real-time and historical air quality data using external APIs
- To preprocess the collected data by handling missing values and scaling features
- To analyze pollution parameters such as PM2.5, PM10, NO_2, SO_2, CO, temperature, and humidity
- To develop and implement machine learning models such as Random Forest and LSTM for AQI prediction
- To predict future AQI values based on past data trends
- To design an interactive and user-friendly dashboard using Streamlit
- To store air quality data and prediction results in a MySQL database for efficient management
- To visualize real-time data, historical trends, and predicted AQI values

3. Feasibility Study

The feasibility study evaluates the practicality and viability of the proposed Air Quality Prediction System. It helps determine whether the system can be successfully developed and implemented using available resources, technology, and within given constraints.

- **Technical Feasibility:** The proposed system is technically feasible as it uses widely available and well-supported technologies such as Python, machine learning libraries (Scikit-learn and TensorFlow), Streamlit for dashboard development, and MySQL for database management.

  The system also utilizes external APIs to fetch real-time and historical air quality data, which simplifies data collection. All required tools and technologies are compatible and can be easily integrated, making the implementation process smooth and efficient.

- **Economic Feasibility:** The system is economically feasible as it is developed using open-source tools and technologies, which reduces development cost. Python and its libraries are free to use Streamlit is open-source MySQL Community Edition is free Many APIs offer free tiers for development Therefore, the overall cost of development and deployment is minimal.

- **Operational Feasibility:** The system is user-friendly and easy to operate. The dashboard provides a simple interface where users can view real-time AQI, historical data, and predicted values without requiring technical knowledge.

  The system can be used by individuals, researchers, and government authorities, making it practical for real-world usage.

4. System Requirements

a. Functional Requirements:

- The system should fetch real-time air quality data using an external API.
- The system should retrieve historical AQI data for analysis.
- The system should preprocess the data by handling missing values and scaling features.
- The system should calculate AQI based on pollutant values such as PM2.5.
- The system should predict future AQI values using machine learning models (Random Forest and LSTM).
- The system should store AQI data and prediction results in a MySQL database.
- The system should allow users to select different locations for AQI monitoring.

b. Non-Functional Requirements:

Usability: The user interface should be simple, interactive, and easy to use.
Reliability: The system should function consistently without failure.
Scalability: The system should handle increasing data and additional locations efficiently.
Security: Database access should be secure and protected from unauthorized access.
Performance: The system should provide quick responses for data fetching and prediction.
Maintainability: The system should be easy to update and maintain.
Accuracy: The prediction models should provide reliable and accurate AQI results.

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c. Hardware Requirements:

Processor: Intel i5 or higher.
RAM: 4 GB minimum (8 GB recommended).
Storage: Minimum 10 GB free disk space.
Internet: Stable high-speed internet connection.

d. Software Requirements

Operating System: Windows / Linux / macOS
Programming Language: Python 3.x
Libraries/Frameworks: Pandas, NumPy, Scikit-learn, TensorFlow/Keras, Plotly, Requests, Joblib
IDE: Visual Studio Code / PyCharm / Jupyter Notebook
Web Framework / Dashboard: Streamlit
Optional Tools: Matplotlib (for visualization), SQLite/MySQL Workbench

5.Proposed System

The proposed Air Quality Prediction System is designed to overcome the limitations of existing air quality monitoring systems by integrating real-time data processing, machine learning-based prediction, and interactive visualization.

The system collects real-time and historical air quality data from external APIs. This data includes various pollution parameters such as PM2.5, PM10, NO_2, SO_2, CO, temperature, and humidity. The collected data is then preprocessed to handle missing values and prepare it for analysis.

Machine learning models such as Random Forest and Long Short-Term Memory (LSTM) are used to analyze patterns in historical data and predict future Air Quality Index (AQI) values. These models help in forecasting air quality in advance, enabling users to take preventive measures.

The system also includes a MySQL database to store location details, pollution data, and prediction results in a structured format. This ensures efficient data management and supports future analysis.

A user-friendly dashboard is developed using Streamlit, which allows users to view real-time AQI data, historical trends, and predicted AQI values. The dashboard also provides visual representations such as charts and gauge indicators for better understanding.

Overall, the proposed system provides an intelligent, efficient, and scalable solution for monitoring and predicting air quality, helping individuals and authorities make informed decisions.

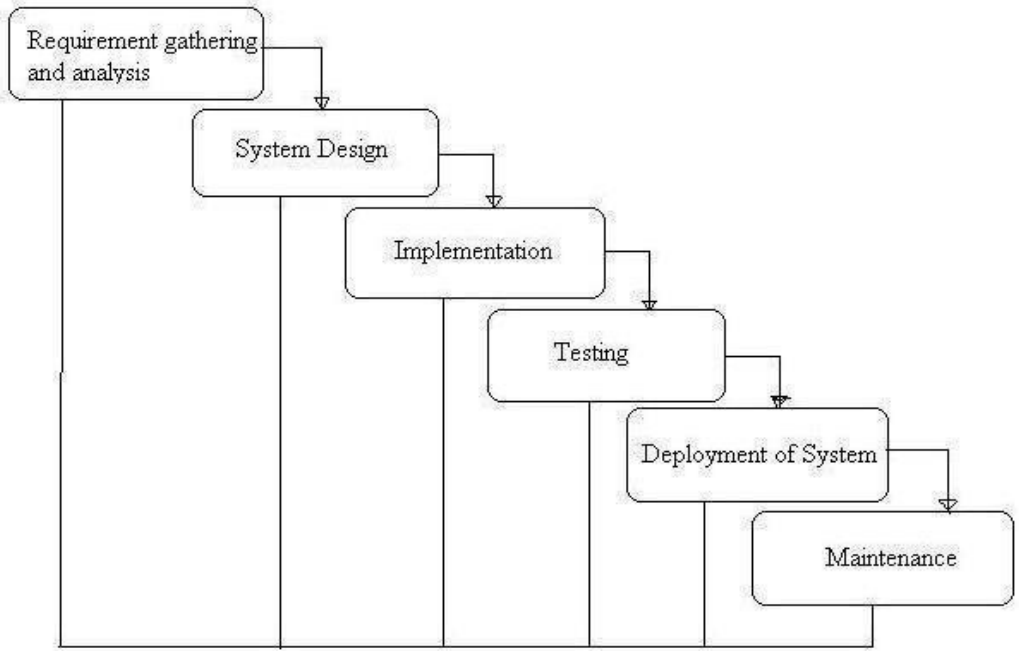

Fig. 5.2 . Phases of Development

6. Phases of Development

i. Requirement Analysis & System Study

- Identify the problem of increasing air pollution and the need for AQI prediction
- Analyze user requirements such as real-time monitoring and future prediction
- Study existing air quality systems and identify their limitations
- Define system requirements for data collection, processing, prediction, and visualization

ii. System Design

- Design the overall architecture of the system, including data flow from API to dashboard
- Define modules such as data collection, preprocessing, model prediction, database, and user interface
- Design the database structure with tables like Location, AQI_Data, and Prediction
- Plan the user interface using Streamlit for displaying AQI data and predictions

iii. Implementation (Coding)

- Develop modules for data collection using API integration
- Implement data preprocessing and machine learning models (Random Forest and LSTM) for AQI prediction
- Integrate MySQL database to store AQI data and prediction results
- Build the user interface using Streamlit to display real-time data and predictions

iv. Testing & Debugging

- Test all modules to ensure correct functionality of data fetching, processing, and prediction
- Validate the accuracy of machine learning models using appropriate evaluation metrics
- Identify and fix errors in API integration, database operations, and user interface
- Ensure the system runs smoothly and handles exceptions or invalid inputs properly

v. Deployment & Maintenance

- Deploy the system using a web-based platform for user access
- Ensure proper integration of all components including API, database, and dashboard
- Monitor system performance and fix issues if they arise
- Update models and system features regularly to improve accuracy and functionality

vi. ER Diagram

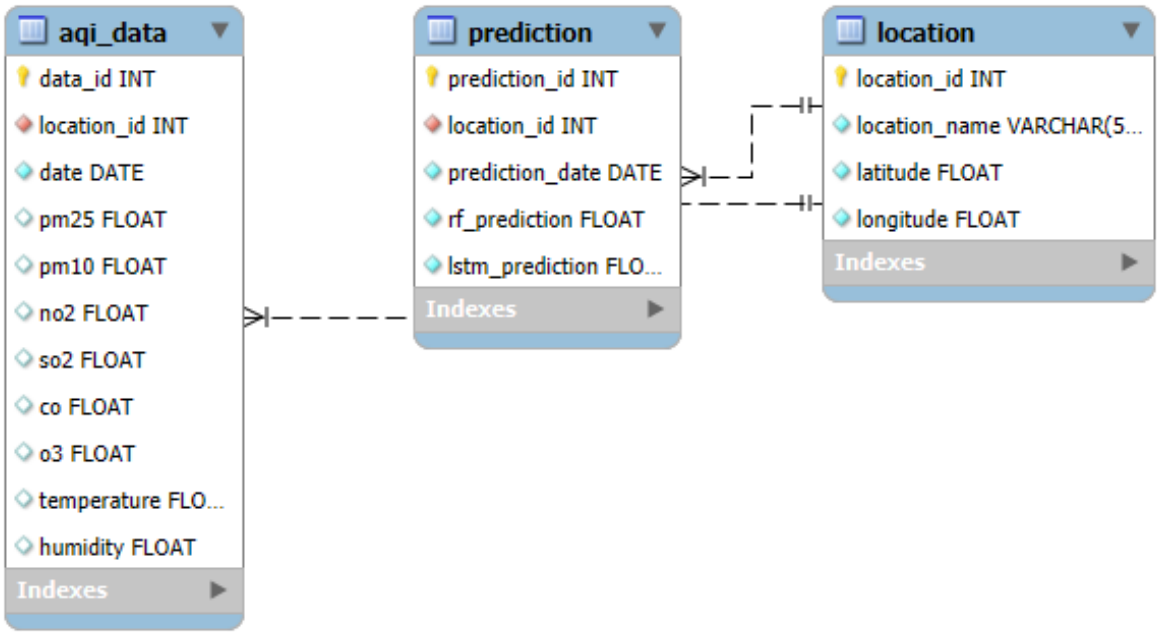

Fig. 2.7. ER Diagram

The ER diagram for the Air Quality Prediction System represents the logical structure of the database, showing entities, their attributes, and relationships among them. It provides a clear view of how locations, air quality data, and prediction results are interconnected within the system.

1. Entities and Their Attributes

Location

Attributes: location_id (PK), location_name, latitude, longitude
Represents different areas for which air quality data is collected and predictions are made.

AQI_Data

Attributes: data_id (PK), location_id (FK), date, pm25, pm10, no2, so2, co, o3, temperature, humidity
Stores detailed air quality parameters for each location along with date-wise pollution data.

Prediction

Attributes: prediction_id (PK), location_id (FK), prediction_date, rf_prediction, lstm_prediction
Stores the predicted AQI values generated by machine learning models for each location.

2. Relationships

Location → AQI_Data:
Each location stores multiple AQI data records collected over time (1 : M).

Location → Prediction:
Each location can have multiple AQI predictions generated for different dates (1 : M).

3. ER Diagram Summary

- The system consists of three main entities: Location, AQI_Data, and Prediction, where Location acts as the central entity storing details of monitoring areas.
- The AQI_Data entity stores historical pollution data for each location, and the Prediction entity stores AQI values predicted by machine learning models, both linked to Location using location_id.
- There is a one-to-many relationship between Location and AQI_Data, and between Location and Prediction, meaning one location can have multiple AQI records and multiple prediction records.

System design

A) Modules

1. Data Collection Module

- Fetches real-time air quality and weather data from the OpenWeather API.
- Collects parameters such as AQI, PM2.5, PM10, NO_2, SO_2, CO, O_3, temperature, and humidity.
- Sends the collected data to the system for storage and further processing.

2. Data Preprocessing Module

- Cleans the data by handling missing or inconsistent values.
- Applies feature scaling to normalize the dataset.
- Prepares input data in the required format for machine learning models.

3. Feature Extraction Module

- Extracts relevant features from raw AQI and weather data such as pollutants, temperature, and humidity.
- Creates additional time-based features like day, month, and previous AQI values.
- Prepares structured input features required for machine learning models.

4. Model Training Module

- Uses historical AQI data to train machine learning models like Random Forest and LSTM.
- Applies preprocessing techniques such as scaling and splitting of training and testing data.
- Optimizes model performance by learning patterns and relationships in the data.

5. Prediction Module

- Takes recent and historical AQI data as input to the trained models.
- Uses Random Forest and LSTM models to generate future AQI predictions.
- Produces output values that represent predicted AQI for the next day.

6.System Module (Backend Services)

- Integrates all components such as data collection, processing, model training, and prediction.
- Manages the overall workflow from input data to final output.
- Displays results on the Streamlit dashboard for user interaction and visualization.

B) Data Structure of All Modules

1.Dataset Structure

The Air Quality Prediction System uses structured relational data stored in a database.

Location Table (location) - Stores information about monitoring areas.

Attributes:

- **location_id** – Unique identifier (Primary Key)
- **location_name** – Name of the location/area
- **latitude** – Latitude coordinate
- **longitude** – Longitude coordinate

AQI_Data Table (aqi_data)

Stores historical air quality and pollution data.

Attributes:

- **data_id** – Unique identifier (Primary Key)
- **location_id** – References Location table (Foreign Key)
- **date** – Date of recorded data
- **pm25** – PM2.5 level
- **pm10** – PM10 level
- **no2** – Nitrogen dioxide level
- **so2** – Sulfur dioxide level
- **co** – Carbon monoxide level
- **o3** – Ozone level
- **temperature** – Temperature value
- **humidity** – Humidity value

Prediction Table (prediction)

Stores AQI values predicted by machine learning models.

Attributes:

- **prediction_id** – Unique identifier (Primary Key)
- **location_id** – References Location table (Foreign Key)
- **prediction_date** – Date of prediction
- **rf_prediction** – AQI predicted using Random Forest
- **lstm_prediction** – AQI predicted using LSTM

2. Relationships Between Data Components

Location → AQI_Data:
Each location can have multiple AQI records collected over time (1 : M).

Location → Prediction:
Each location can have multiple predicted AQI records for different dates (1 : M).

This structure ensures efficient data processing, reduces redundancy, and maintains data integrity. It supports scalable machine learning workflows and enables fast and reliable AQI prediction.

Foreign Keys:

Location Table (location)

- Used as the primary reference table for the system.

AQI_Data Table (aqi_data)

- **location_id → References location(location_id)**
  Establishes a relationship between AQI_Data and Location, indicating that each AQI record belongs to a specific location.

Prediction Table (prediction)

- **location_id → References location(location_id)**
  Establishes a relationship between Prediction and Location, indicating that each predicted AQI record is associated with a specific location.

3.Relationships Between Tables (ER/Class Diagram)

Location → AQI_Data:
One location can have multiple AQI data records collected over time (1 : M).
Location → Prediction:
One location can have multiple AQI prediction records for different dates (1 : M).
AQI_Data → Prediction (Conceptual Relationship):
Historical AQI data is used as input for machine learning models to generate predictions.

C) Screenshots

Fig. 3.1.AQI Prediction System Dashboard

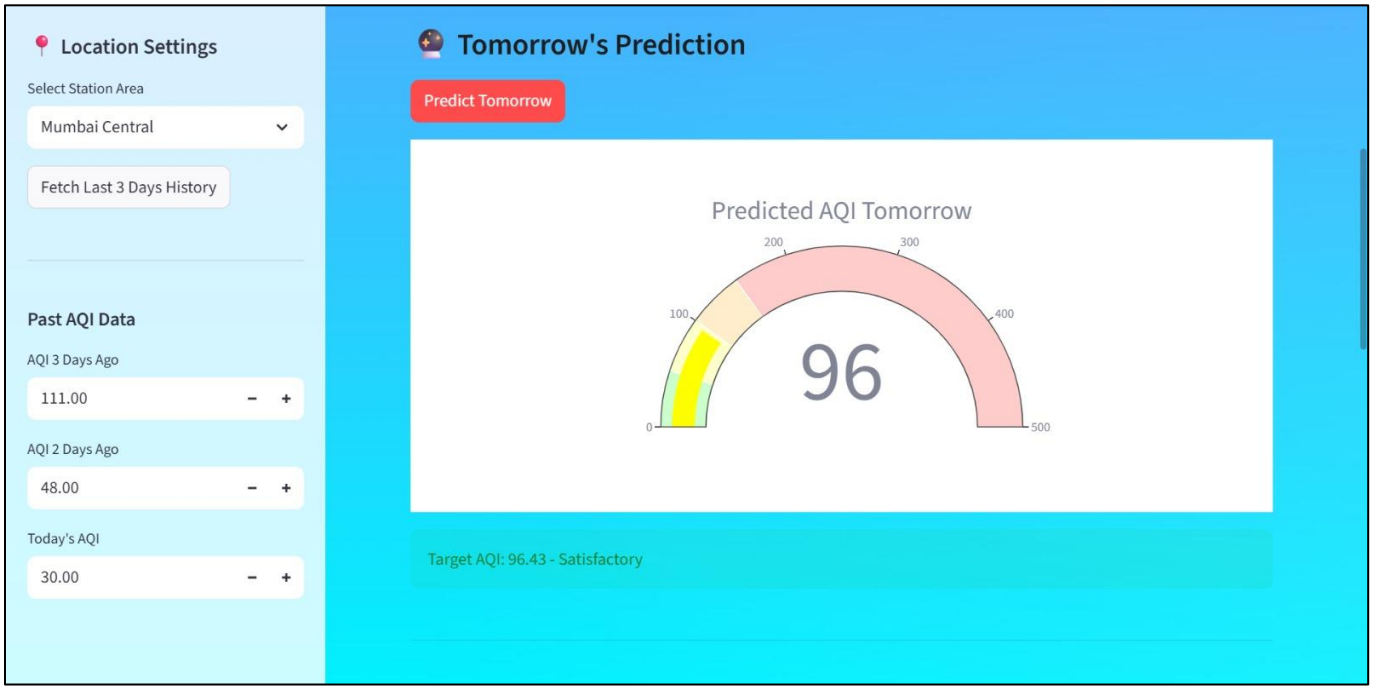

Fig. 3.2. Tomorrow's AQI Predection

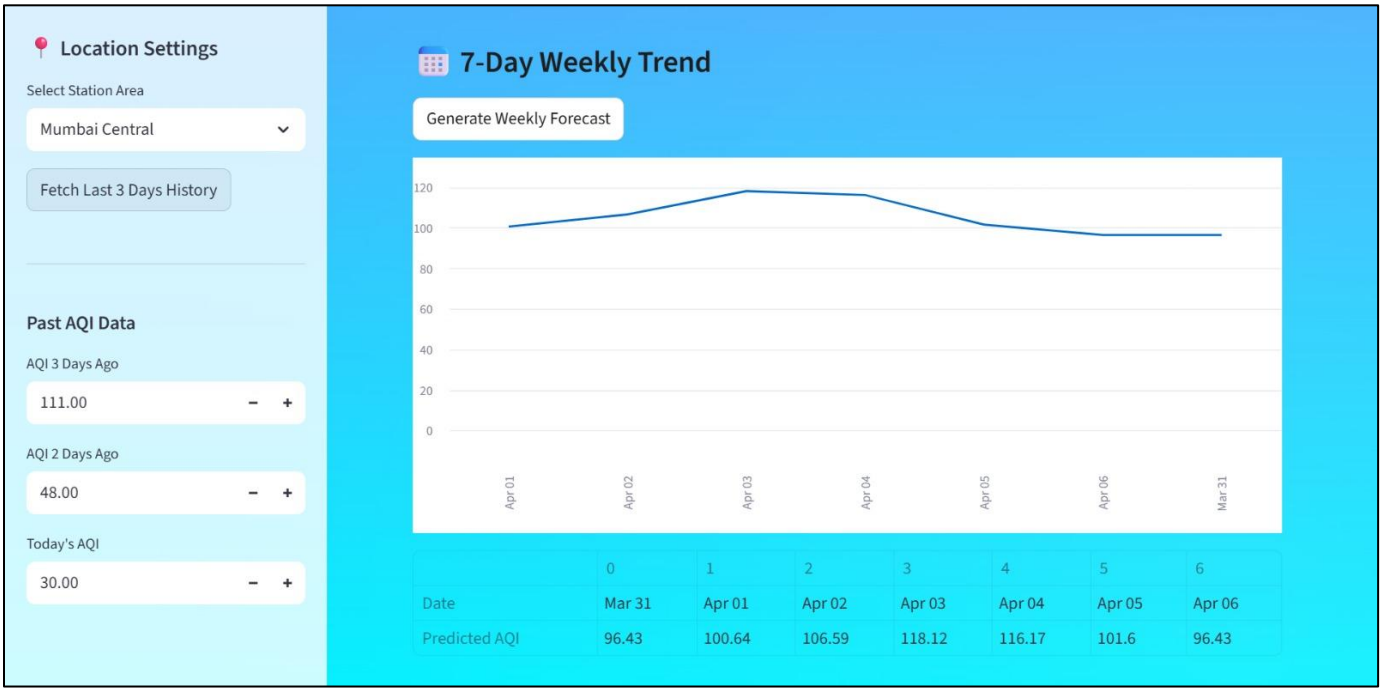

Fig. 3.3. 7 Days Weekly Forecast

Fig. 3.4. Air Quality Index (AQI) Scale

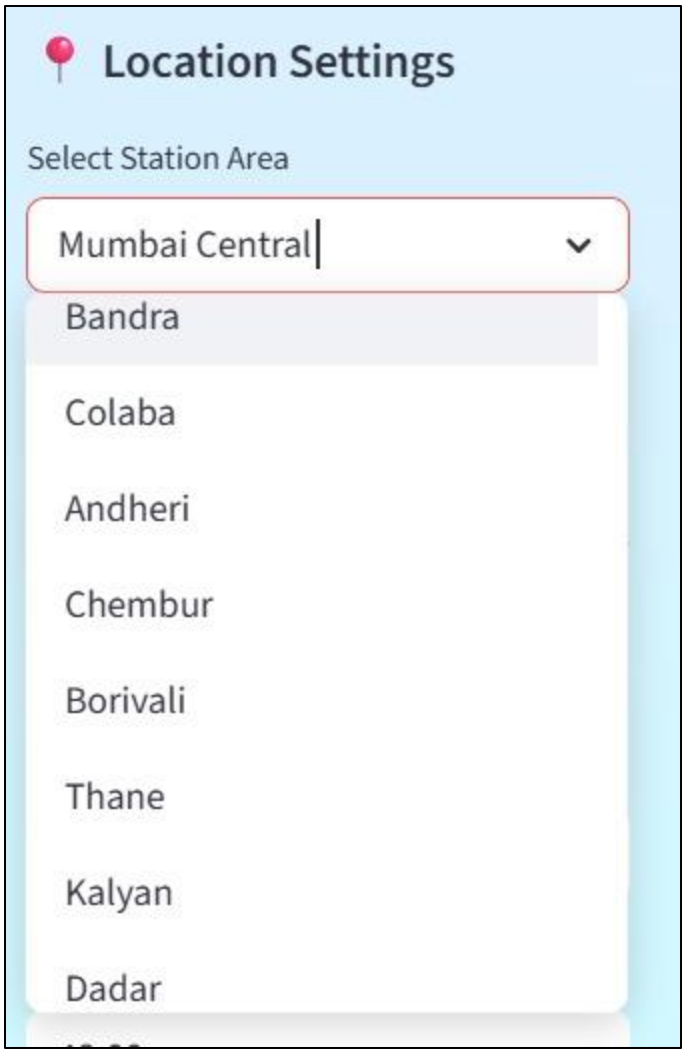

Fig. 3.5. Mumbai's Station Area

Implementation

The implementation phase involves converting the system design into a functional software application. The Air Quality Prediction System is developed using Python, machine learning techniques, and real-time data integration for AQI forecasting. This section describes the technologies, logic, and design used in the implementation.

1. Technology Stack

- **Programming Language:** Python (3.x)
- **Libraries/Frameworks:** Pandas, NumPy, Scikit-learn, TensorFlow/Keras
- **Machine Learning Models:** Random Forest and LSTM
- **API Integration:** OpenWeather API for real-time air quality and weather data
- **Database:** SQLite for storing AQI data and predictions
- **Web Framework / Dashboard:** Streamlit for interactive visualization
- **Development Environment:** Jupyter Notebook
- **Data Source:** OpenWeather API and historical AQI data

2. Backend Implementation

The backend handles all core functionalities including data collection, preprocessing, model prediction, and database operations. It is responsible for processing real-time data and generating accurate AQI predictions.

Key Functionalities:

• **Data Collection:**
Fetching real-time and historical air quality data using the OpenWeather API, including parameters such as AQI, PM2.5, PM10, temperature, humidity, and other pollutants.

• **Data Preprocessing:**
Cleaning and preparing the data by handling missing values and applying feature scaling using a scaler to normalize input values for machine learning models.

• **Model Prediction:**
Using machine learning models such as Random Forest and LSTM to analyze historical data and predict future AQI values.

• **7-Day Forecasting:**
Generating future AQI predictions for the next 7 days using a recursive prediction approach.

• **Model Evaluation:**
Ensuring prediction accuracy by evaluating models using metrics such as RMSE (Root Mean Square Error) and MAE (Mean Absolute Error).

3. User Interface Implementation

The system is implemented as a web-based interface using Streamlit to provide an interactive and user-friendly dashboard.

- The user selects a location from the sidebar to view air quality data.
- The user selects a location from the sidebar to view air quality data.
- The data is displayed on the dashboard, including AQI, temperature, humidity, and pollutant levels.
- The user can click a button to predict the next day's AQI using machine learning models.
- The user can click a button to predict the next day's AQI using machine learning models.

4. Dataset Implementation

The system uses a structured dataset containing real-time and historical air quality data collected from external APIs.

Dataset Components:

- Location: Name of the area
- Date/Time: Timestamp of data collection
- AQI: Air Quality Index value
- PM2.5, PM10: Particulate matter levels
- NO_2, SO_2, CO, O_3: Pollutant concentrations
- Temperature: Atmospheric temperature
- Humidity: Humidity level

Processing Steps:

- Humidity: Humidity level
- Handle missing values and clean the dataset
- Apply feature scaling for normalization
- Apply feature scaling for normalization

5. Security Implementation

• **Data Validation:**
Ensures that the data fetched from the API is valid and not empty before processing and prediction.

• **Model Integrity:**
Trained machine learning models are securely stored using serialization techniques such as joblib and TensorFlow model saving formats.

• **Error Handling:**
Handles exceptions such as API failures, missing data, and model loading errors to ensure smooth system.

- **Input Sanitization:**
  Validates user inputs such as AQI values and location selection to prevent incorrect data from affecting predictions.

- **Performance Optimization:**
  Ensures efficient processing of data and fast prediction response time for a smooth user experience.

Results and Discussion

The development and implementation of the Air Quality Prediction System produced several notable outcomes. This section highlights the system's functional effectiveness, performance, and areas for improvement based on testing and execution.

1.Results

Functional Results

- **Data Collection Module:**

  Real-time and historical air quality data are successfully fetched using the API. The system efficiently handles multiple locations and large volumes of environmental data.

- **Data Preprocessing Module:**

  The collected data is cleaned and processed by handling missing values and applying feature scaling. This ensures high-quality input for machine learning models.

- **Feature Processing Module:**

  Environmental parameters such as AQI, PM2.5, PM10, temperature, and humidity are converted into structured numerical formats suitable for model prediction.

- **Model Prediction Module:**

  Machine learning models such as Random Forest and LSTM are used effectively to predict AQI values based on historical data patterns.

- **Prediction Output Module:**

  The system successfully predicts next-day AQI and generates a 7-day forecast. The predictions are displayed quickly with good accuracy.

- **Model Evaluation:**

  The performance of the models is evaluated using metrics such as RMSE and MAE, ensuring reliable and accurate predictions.

Non-Functional Results

- **Performance:**

  The system provides fast response time while fetching real-time data and generating AQI predictions, ensuring smooth user interaction.

- **Usability:**

  The Streamlit-based dashboard is simple, interactive, and easy to use, allowing users to access information without technical knowledge.

- **Reliability:**

  The system consistently fetches data, performs predictions, and handles errors efficiently, ensuring stable performance.

- **Scalability:**

  The system can be extended to support more locations, additional features, and larger datasets without major changes.

- **Maintainability:**

  The modular structure of the code (API, model, database, UI) makes it easy to update, debug, and enhance the system.

- **Security:**

  Basic validation and error handling ensure safe data processing and prevent system failures.

2.Discussion

a. Achievement of Objectives

- All major objectives—such as real-time data collection, AQI prediction, and dashboard visualization—were successfully achieved.
- Integration of API, machine learning models, database, and user interface was implemented effectively.

b. User Experience

- The Streamlit dashboard provides a simple, interactive, and user-friendly interface.
- Features like location selection, real-time updates, and prediction buttons make the system easy to use and understand.

c. Challenges Faced

- Handling real-time API data and managing missing or inconsistent values required additional preprocessing.
- Training and integrating machine learning models like LSTM required proper data formatting and tuning.
- Ensuring smooth integration between API, database, and dashboard required multiple testing iterations.

d. Limitations

- The system depends on external API data, which may sometimes be unavailable or inaccurate.
- Prediction accuracy may vary due to limited historical data.
- Currently supports limited locations and basic features.

e. Future Enhancements

- Expand the system to support more locations and larger datasets.
- Improve prediction accuracy using advanced models and more data.
- Improve prediction accuracy using advanced models and more data.
- Enhance visualization with more advanced dashboards and analytics.

Conclusion

The development of the Air Quality Prediction System has successfully demonstrated the design and implementation of an intelligent, efficient, and reliable platform for monitoring and predicting air quality using machine learning techniques. The system enables users to:

- Monitor real-time air quality data effectively.
- Analyze environmental parameters such as AQI, temperature, humidity, and pollutants.

Key achievements include:

- Successful implementation of machine learning algorithms such as Random Forest and LSTM for accurate AQI prediction.
- Integration of real-time data using API and storage using SQLite database.
- A modular architecture that improves scalability and allows future enhancements.
- Effective visualization using an interactive dashboard with charts and graphs.
- Efficient handling of data and prediction processes, ensuring quick and reliable results.

Overall, the Air Quality Prediction System provides a robust and intelligent solution for air quality monitoring and forecasting. It helps users and authorities take preventive measures, improves environmental awareness, and serves as a strong foundation for future enhancements such as advanced prediction models, alert systems, and large-scale deployment.

Future scope

The Air Quality Prediction System can be further enhanced by incorporating advanced features and technologies to improve its accuracy, usability, and real-world impact.

- **Advanced Machine Learning Models:**

  Future versions can use deep learning models such as GRU or hybrid models to improve prediction accuracy.

- **More Locations Coverage:**

  The system can be expanded to support multiple cities and countries for large-scale air quality monitoring.

- **Real-Time Alerts & Notifications:**

  Develop an Android/iOS application for better accessibility and user convenience.

- **Integration with IoT Sensors:**

  Connect with real-time air quality sensors for more accurate and localized data collection.

- **Government & Smart City Integration:**

  The system can be integrated with smart city infrastructure for better environmental planning and pollution control.

- **Advanced Data Visualization:**

  Include heatmaps, trend analysis, and comparative dashboards for better insights.

- **Health Recommendation System:**

  Provide personalized health suggestions based on AQI levels (e.g., mask usage, outdoor activity limits).

Bibliography and References

In preparing this project report on the Air Quality Prediction System, several books, research papers, and online resources were referred to. These materials provided valuable insights into machine learning, air quality analysis, data preprocessing, and system development methodologies.

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